

Ahnaf Shahriar

[Email](#) | [LinkedIn](#) | [Github](#)

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2021 – May 2026

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses:** Digital Communication Systems, Computer Networks, Computer Architecture, Real Time Operating Systems, Digital Hardware systems(Verilog)

EXPERIENCE

AI HW Engineering Intern

Aug. 2025 – Dec. 2025

Tesla

Palo Alto, CA

- **Emulation bringup:** Led GPU bringup in *ZeBu* emulation with *Zephyr ZTest* for firmware integration testing
- **Rust Development:** Built and Linux Rust drivers for Power Supply automations and monitoring for 20+ PSU units
- **Post-silicon validation:** Designed automated *Python* validation scripts that exercises FLIT *PCIe* TLP and LTSSM testing
- **Performance & power characterization:** Profiled end-to-end Camera workloads and implemented tests to characterize performance and power during early bringup.

Undergraduate Research Assistant

May. 2025 – Aug. 2025

University of Waterloo

Waterloo, ON

- **Research:** Conducting Case studies on Static Analysis type checking for Java.
- **Compiler Optimization:** Implementing bytecode storage on Nullness checker for EISOP framework.

Wireless Firmware Developer

Jan. 2025 – Apr. 2025

ON Semiconductor Corporation

Waterloo, ON

- **SDK Development:** Integrated *I3C* support and *BLE* profile services into sample code.
- **Wireless Connectivity:** Developed and integrated cutting-edge *LE Audio*, *HFP*, and *A2DP* functionality into an all-in-one Application.
- **Wireless Manager:** Modularized and implemented a state machine for power efficiency by 25%
- **Test Automation:** Automated SDK test cases, reducing manual testing efforts by up to 80%

IC Design and Verification Intern

Jan. 2024 – May 2024

NXP Semiconductors Canada

Kanata, ON

- **IP Design:** Developed multiple IP blocks for TCAM and Ethernet Parsing for NoC upto 100Gbps.
- **Timing Analysis:** Spearheaded critical path improvements for IP to meet 600Mhz from 400Mhz.
- **Functional Testing:** Designed brand new End-to-End functional tests for *ECC Detection* on Chip.

IC Design Verification Intern

May 2023 – Aug. 2023

NXP Semiconductors Canada

Kanata, ON

- **UVM SystemVerilog:** Designed the *IP specific Virtual Sequence* to complete corresponding IP coverage exceeding 95%.
- **Test Planning:** Created Simulation scenarios for testing features in High speed Dataplane processing.
- **Simulation Testing:** Created Simulation scenarios for testing features in High speed Dataplane processing.

PROJECTS

RISC-V processor: A 5 stage pipelined FPGA processor in Verilog. Designed and tested with Vivado for Zynq-7000.

Stereo System: An FPGA designed in Quartus and programmed in C. Implements stereo playback speed options.

Real Time Executable: A STM32 RTOS implementation capable of Pre-emptive task switching and its very own Malloc.

VHDL Compiler: A Java Compiler for creating combinational VHDL circuits. Using a boolean intermediate representation

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Tcl, Perl and Bash scripting, ASM, VHDL, SystemVerilog/Verilog

Tools: Quartus, Git, Linux, Qemu, WireShark, UVM, Matlab, Synopsys DC, Cadence Xcelium, Vivado

Hardware: Oscilloscopes, Logic Analyzer, Multimeters, Spectrum Analyzer

Protocols: TCP/IP, JTAG, Serial, Ethernet, CAN/CAN-FD, LIN, AXI